# DINDAKURTHI KUSHAL

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## SUMMARY

AI & ML graduate with hands-on experience in building enterprise search systems, developing software applications, and working with Python, backend technologies, and web frameworks. Brings a strong foundation in problem-solving, ownership, and team collaboration, with an interest in applying technical skills to real-world challenges and growing through meaningful professional opportunities.

## SKILLS

**Languages & Frameworks:** Python, Java, SQL, JavaScript, FastAPI, Flask, React, Streamlit, HTML, CSS

**Agentic AI:** RAG, Azure AI Foundry, LangGraph, Azure OpenAI, Hybrid Search (BM25 + Dense), Reciprocal Rank Fusion, Semantic Search, Query Expansion, Embeddings (Sentence Transformers, BGE), Qdrant, Ollama, Reranking, Jupyter Notebook

**Integration & Backend:** Microsoft Graph API, Confluence REST API, SharePoint, Microsoft Teams Bot, REST APIs, Git, VS Code

## EXPERIENCE

**AI Intern** | Lumen Technologies | Bengaluru *Feb 2026 - Aug 2026*

- Architected an end-to-end enterprise search assistant, providing company IT staff with a natural language search interface, and owned its delivery through production deployment with regular mentor and manager reviews.
- Built an on premises search system using Python, FastAPI and Qdrant, indexing ∼60,000 documents and ∼700 tickets across Confluence, Teams, SharePoint and CSV exports, reducing mean time to answer by ∼40%.
- Implemented hybrid retrieval using parallel BM25 and vector search (all-MiniLM-L6-v2), combining results via Reciprocal Rank Fusion with title, heading and phrase boosts to rank exact matches over generic keyword hits.
- Moved from an initial reranking approach to RRF with metadata boosting to resolve BM25/vector score-scale mismatch; validated ranking on sample queries (error codes, page titles, section phrases) with the expected result returned to the top results.
- Designed heading aware chunking with token overlap and heading path prefixes, and a BeautifulSoup extractor that preserves Confluence heading hierarchy, then collapsed chunk hits into page level results with provenance tags (BM25, vector or both).
- Developed ingestion connectors for Confluence (full page body) and Microsoft Graph (Teams and SharePoint file title and folder path indexing), with YAML exclusion rules, incremental change detected sync scheduled via APScheduler to incorporate new and updated content, and cache backed restarts in under 5 seconds. Exposed search through FastAPI endpoints consumed by a Streamlit UI and a Microsoft Teams bot that returns the top results as Adaptive Cards.
- Integrated Llama 3.1 through Ollama for query expansion with a rule-based synonym fallback on timeout or parse failure, keeping search available when the LLM is slow or unavailable; added regex based PII masking (emails, phone numbers, IPs) before embedding so sensitive data never enters the vector index, meeting on-premises privacy requirements.

**Chief Video Editor** | e-DAM | Hyderabad *Apr 2024 - Mar 2025*

- Led the club's video team, delivering 50+ videos in 12 months by assigning tasks, setting priorities and tracking deadlines, while reviewing every video, providing actionable feedback and maintaining consistent quality standards across all deliverables.
- Coordinated with club members and event organizers to understand requirements, align on timelines and adapt to last-minute changes. Took end-to-end ownership of delivery, from planning and task allocation to final approval, building skills in team leadership, communication and working under deadlines.

## EDUCATION

**Institute of Aeronautical Engineering**, Hyderabad 2022 - 2026

B.Tech in Computer Science and Engineering (AI & ML) GPA: 8.44/10

**Coursework:** Object-Oriented Software Engineering, Data Structures, Machine Learning, Artificial Intelligence, Database Management Systems, Computer Vision, Natural Language Processing, Operating Systems, Computer Networks

## PROJECTS

**CardioInsight - Web-Based Cardiovascular Disease Prediction System** | GitHub *Jan 2025 - Apr 2025*

*Python, SVM, XGBoost, CatBoost, Pandas, NumPy, Flask, React*

- Engineered a full stack ML application to predict cardiovascular disease risk from clinical data, addressing the delay and cost of traditional multi test diagnosis with instant, accessible risk screening through a Flask API and React interface. Users enter health details and receive a risk result with personalized health advice, performance visualizations and a downloadable PDF report.
- Designed a stacking ensemble (XGBoost, CatBoost and SVM base learners with a Random Forest meta-learner) trained on a 2,181 record Kaggle clinical dataset, reaching 94% accuracy versus 91% for the best single model (XGBoost) and 67% for SVM.
- Evaluated the model on an 80/20 train-test split using accuracy, precision, recall, F1 score and a confusion matrix.

**Gesture Music System** | GitHub *Aug 2024 - Nov 2024*

*Python, OpenCV, Flask*

- Built a touch free music controller in Python with OpenCV to capture webcam video, detect hands in real time and recognize gestures, enabling music control without physical input devices.
- Mapped recognized gestures to playback controls and integrated the recognition module with a Flask backend and responsive web interface to trigger music actions end to end.
- Tested and refined detection across different hand positions and lighting conditions to improve recognition consistency.

**Restaurant Management System** | GitHub *Mar 2024 - Jun 2024*

*HTML, CSS, JavaScript*

- Developed a responsive restaurant management web application in HTML, CSS and JavaScript that brings menu browsing, table reservations and food ordering into a single interface.
- Implemented reservation forms with JavaScript input validation and an interactive ordering flow that updates without reloading; designed responsive layouts for desktop and mobile screens.

## CERTIFICATIONS

Salesforce AI Specialist
The Joy of Computing Using Python
Programming Essentials in Python